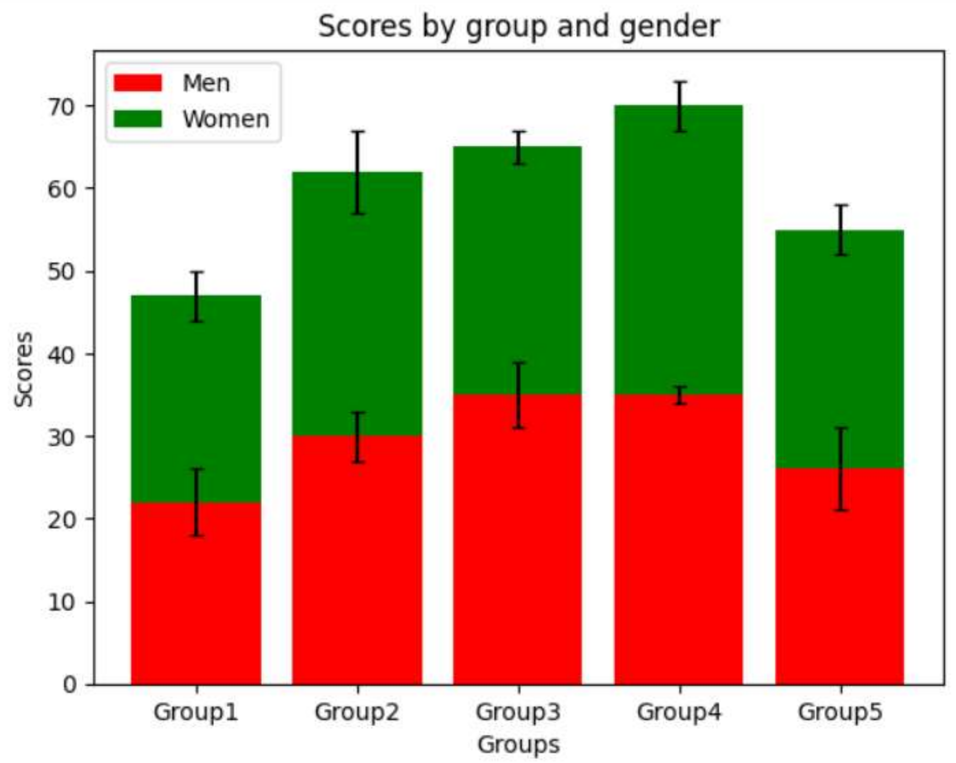
LAB EXPERIMENTS 30-32

30.CODING:

```
import numpy as np
import matplotlib.pyplot as plt

groups = ['Group1', 'Group2', 'Group3', 'Group4', 'Group5']

men = [22, 30, 35, 35, 26]
women = [25, 32, 30, 35, 29]

men_std = [4, 3, 4, 1, 5]
women_std = [3, 5, 2, 3, 3]

x = np.arange(len(groups))

plt.bar(x, men, yerr=men_std, label='Men', color='red', capsize=3)
plt.bar(x, women, bottom=men, yerr=women_std,
        label='Women', color='green', capsize=3)

plt.xlabel('Groups')
plt.ylabel('Scores')
plt.title('Scores by group and gender')
plt.xticks(x, groups)
plt.legend()

plt.show()
```

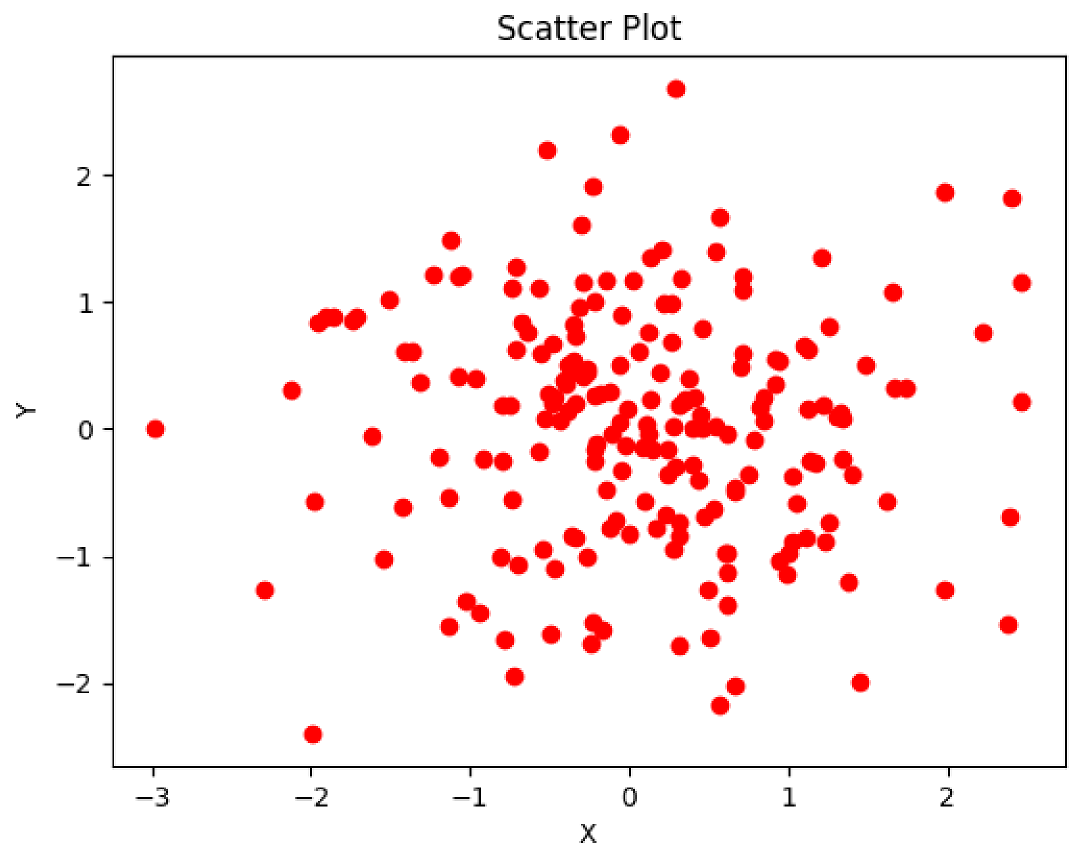
OUTPUT:



31.CODING:

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(10)

x = np.random.randn(200)
y = np.random.randn(200)

plt.scatter(x, y, color='red')

plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot')

plt.show()
```

OUTPUT:



32.CODING:

```
import numpy as np
import matplotlib.pyplot as plt


np.random.seed(10)


x = np.random.randn(100)
y = np.random.randn(100)


plt.scatter(x, y, facecolors='none', edgecolors='green')


plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot with Empty Circles')
```

```
plt.show()
```

OUTPUT: